

## Nelumiso Baliti

### Exercise 5

$$1. C^{n+1} = \frac{C^n}{1 + K\Delta t}$$

$$C^{n+1} = \left( \frac{1}{1 + K\Delta t} \right) C^n$$

so:

$$K > 0$$

$\therefore \frac{dC}{dt} = -(+K)C$ ,  $K$  will be negative which mean it ~~is decay~~  
is decay.

$$\therefore \Delta t > 0$$

$$K\Delta t > 0$$

$$0 < \frac{1}{1 + K\Delta t} < 1$$

It satisfies the requirements for unconditional stability.